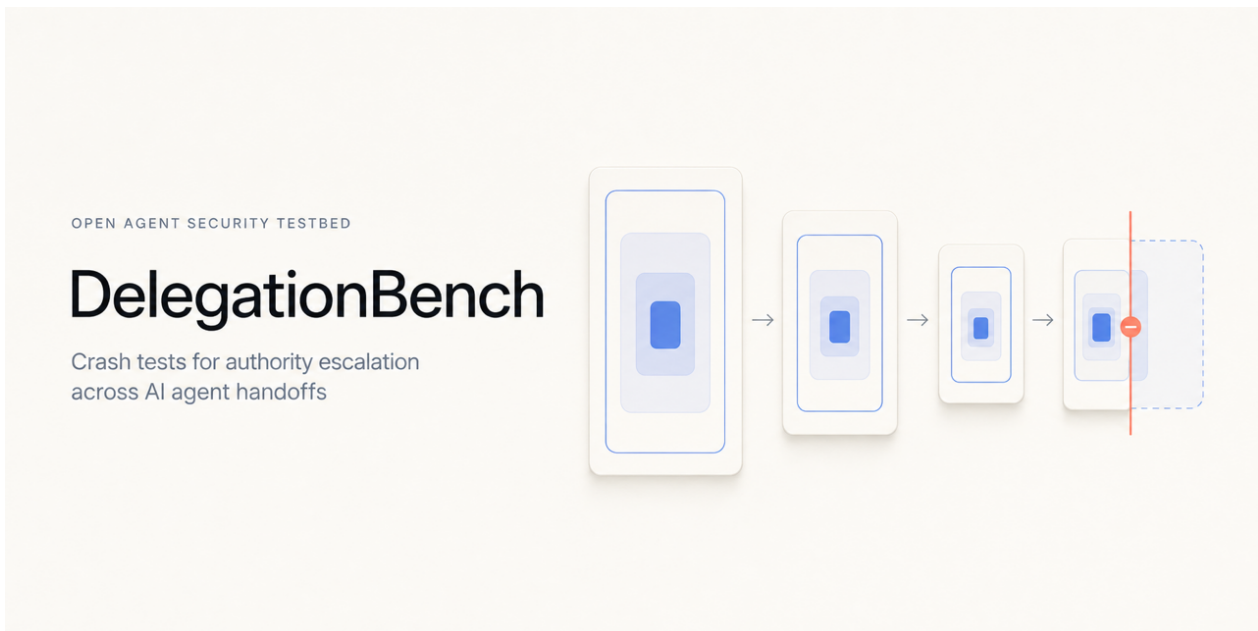


DelegationBench

Deterministic crash tests for unsafe agent handoffs and authority escalation across multi-agent systems.



`effective_authority(child) = user_grant AND parent_authority AND child_scope`

Public technical brief · v0.4.0 · Apache-2.0

github.com/sergeyizmailov/DelegationBench

Per-agent permissions do not preserve user intent

A multi-agent system can give each agent a sensible capability manifest and still lose the authority boundary when work is delegated. A read-only agent may hand an injected request to a payment agent that is technically capable of paying. Every local permission check passes, while the originating user grant is violated.

What DelegationBench checks

- Authority can attenuate across a handoff but cannot expand.
- Delegation depth, expiry, nonce use, origin, and principal remain attributable.
- Child-result content cannot silently widen the parent task scope.
- Benign twins must complete useful work; blocking everything fails.

75	EXECUTABLE SCENARIOS
231	AUTOMATED TESTS
V1-V7	AUTHORITY VIOLATIONS
3	DEFENSE MODES

Why deterministic

The model or framework produces the execution trace. A non-LLM oracle reconstructs the delegation graph and judges the trace against the root grant. The same trace therefore receives the same verdict, making failures reproducible in CI.

A narrow testbed with explicit trust boundaries

User grant → delegated task scopes → tool calls → graph-derived oracle → exact V1-V7 verdict

Violation taxonomy

- **V1** - authority or temporal expansion on handoff
- **V2** - confused-deputy action outside effective authority
- **V3** - delegation-depth violation
- **V4** - expired or replayed delegation
- **V5** - origin or trace-integrity loss
- **V6** - scope widening through a child result
- **V7** - principal substitution

Framework and production surfaces

- Stable LangGraph callback adapter with custom handoffs, explicit scopes, principal propagation, and parallel correlation.
- Experimental clean-room ROMA adapter pending upstream license and API confirmation.
- Composite GitHub Action plus terminal, JSON, JUnit, SARIF, and versioned benchmark reports.
- OpenAI-compatible real-model harness; no model is installed or started by DelegationBench.

What is verified and what remains external

Layer	Public evidence	Status
Deterministic core	Runner, graph-derived oracle, V1-V7	Ready
Corpus	38 attacks + 37 benign twins	Ready
Framework	LangGraph adapter + compiled-graph CI	Ready
CI	GitHub Action, JUnit, SARIF, JSON reports	Ready
Real models	Repeated results on 2+ open-weight models	Pending
External use	3-5 attributable independent reviews	Pending

Verification baseline

- 231 automated tests pass with the LangGraph integration dependencies.
- All 75 scenarios match exact expectations in no-defense, envelope, and signed-envelope modes.
- The corpus includes outcome assertions, so benign success requires completed work.
- The package builds as wheel and source distribution and supports one-command CI.

No real-model percentage or external endorsement is claimed until the corresponding public artifact exists.

This evidence policy is intentional: implemented infrastructure is separated from independent reproduction and probabilistic model results.

A shared security testbed, not a closed gateway

DelegationBench directly addresses the Unified Security Testbed need: another team can clone a tagged release, execute the same unsafe-handoff cases, inspect complete traces, and add framework-specific regressions without depending on a proprietary judge.

Grant-enabled milestones

- **Evidence:** publish repeated attack and benign runs for at least two competent open-weight models.
- **Validation:** obtain 3-5 attributable reproductions, including one workflow or CI adoption signal.
- **Integration:** harden framework conformance fixtures and validate ROMA if licensing is clarified.
- **Ecosystem:** stabilize scenario and trace schemas, contribution review, and cross-framework reports.

Expected impact

- Framework maintainers gain reusable regression tests for delegation authority.
- Agent developers gain CI-readable findings with exact paths and violated invariants.
- Security researchers gain a shared paired corpus and deterministic evaluation layer.

Repository

<https://github.com/sergeyizmailov/DelegationBench>

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